

2LA, Test 1 (17 May 2021)

①

$$\begin{pmatrix} 0 & 4 & -3 \\ -4 & 0 & 7 \\ 3 & -7 & 0 \end{pmatrix} \checkmark \checkmark$$

(-1pt per error)

2pt total

②

(a) $A_2 \checkmark$

(b) $A_4 \checkmark$

(c) $A_3 \checkmark$

(d) $A_2 \checkmark$

(e) $A_4 \checkmark \checkmark$

5pt total

③ (a)

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 0 & 1 \end{pmatrix}$$

$$E_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{pmatrix} \checkmark$$

(one mark only if both correct)

$$E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1/3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$E_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \checkmark$$

(one mark only if both correct)

$$E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$E_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \checkmark$$

(one mark only if both correct)

$$(b) \quad A = E_1^{-1} E_2^{-1} E_3^{-1} \begin{pmatrix} 1 & 0 & -5 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & -5 \\ 0 & 0 & -3 \\ -5 & 4 & 25 \end{pmatrix} \checkmark \checkmark \checkmark \checkmark$$

-1pt per error

final answer
is sufficient

④ We need two elementary operations to reduce $\begin{pmatrix} 1 & 0 & -5 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

to I_3 .

We take

$$e_4: R_1 \rightarrow R_1 + 5R_3$$

$$e_4 \left(\begin{pmatrix} 1 & 0 & -5 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$e_5: R_2 \rightarrow \frac{1}{4} R_2$$

$$e_5 \left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

Hence with

$$E_4 = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \checkmark$$

and

$$E_5 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix} \checkmark$$

we have

$$E_5 E_4 E_3 E_2 E_1 A = I_3.$$

We see that A is invertible since A is row equivalent to I_3 .

we get

$$A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} E_5^{-1},$$

where E_1, E_2 , and E_3 are the elementary matrices given in Q3(a).

(e_4 and e_5 (and hence E_4 and E_5) may be swapped.)

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$$\begin{pmatrix} \textcircled{1} & -6 & 3 & 3 & | & 9 \\ 0 & \textcircled{2} & 0 & -1 & | & 3 \\ 0 & 4 & 0 & -2 & | & 6 \\ 0 & 0 & -1 & 6 & | & 1 \\ 0 & 0 & k & k^2 & | & 6 \end{pmatrix} \quad R_3 \rightarrow R_3 - 2R_2$$

$$\begin{pmatrix} \textcircled{1} & -6 & 3 & 3 & | & 9 \\ 0 & \textcircled{2} & 0 & -1 & | & 3 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & 6 & | & 1 \\ 0 & 0 & k & k^2 & | & 6 \end{pmatrix} \quad R_3 \leftrightarrow R_4$$

$$\begin{pmatrix} \textcircled{1} & -6 & 3 & 3 & | & 9 \\ 0 & \textcircled{2} & 0 & -1 & | & 3 \\ 0 & 0 & \textcircled{-1} & 6 & | & 1 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & k & k^2 & | & 6 \end{pmatrix} \quad R_4 \leftrightarrow R_5$$

$$\begin{pmatrix} \textcircled{1} & -6 & 3 & 3 & | & 9 \\ 0 & \textcircled{2} & 0 & -1 & | & 3 \\ 0 & 0 & \textcircled{-1} & 6 & | & 1 \\ 0 & 0 & k & k^2 & | & 6 \\ 0 & 0 & 0 & 0 & | & 0 \end{pmatrix} \quad R_4 \rightarrow R_4 + kR_3$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 3 & 9 \\ 0 & 2 & 0 & -1 & 3 \\ 0 & 0 & -1 & 6 & 1 \\ 0 & 0 & 0 & K^2+6K & 6+K \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad \checkmark \checkmark$$

Of course, there are many other ways of determining a row-echelon form.

(a) "No solution" is possible only if there is a row
 $"0 \ 0 \ 0 \ 0 \mid \neq 0"$.

Hence we need

$$K^2+6K=0 \text{ and } 6+K \neq 0.$$

Therefore,

$$\underline{\underline{K=0}} \quad \checkmark$$

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(b) Infinitely many solutions iff row 4 in the row-echelon form is

$$[0 \ 0 \ 0 \ 0 \mid 0]$$

Hence

$$K^2 + 6K = 0 \text{ and } 6 + K = 0.$$

It follows that

$$\underline{\underline{K = -6}}$$

Let's determine the solutions for $K = -6$.

For $K = -6$, we have only three non-zero rows:

$$\begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ \textcircled{1} & -6 & 3 & 3 & \mid & 9 \\ 0 & \textcircled{2} & 0 & -1 & \mid & 3 \\ 0 & 0 & \textcircled{-1} & 6 & \mid & 1 \end{pmatrix}$$

x_4 free variable

We set

$$\underline{x_4 = t}, \quad t \in \mathbb{R}.$$

$$-x_3 + 6x_4 = 1$$

$$-x_3 + 6t = 1$$

$$\underline{x_3 = 6t - 1}$$

$$2x_2 - x_4 = 3$$

$$2x_2 - t = 3$$

$$\underline{x_2 = \frac{3}{2} + \frac{1}{2}t}$$

$$x_1 - 6x_2 + 3x_3 + 3x_4 = 9$$

$$x_1 - 6\left(\frac{3}{2} + \frac{1}{2}t\right) + 3(6t - 1) + 3t = 9$$

$$x_1 - 9 - \cancel{3t} + 18t - 3 + \cancel{3t} = 9$$

$$x_1 + 18t - 12 = 9$$

$$\underline{x_1 = 21 - 18t}$$

✓✓

Hence

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 21 \\ 3/2 \\ -1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -12 \\ 1/2 \\ 6 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}$$


(c) Unique solution iff neither
(a) nor (b).

Hence iff $K \neq 0$ and $K \neq -6$.

⑥ (a) Converse:

For every square matrix A ,
if A is non-singular then $A^T A = I$.

✓

(b) Contrapositive:

For every square matrix A ,
if A is singular then $A^T A \neq I$.

✓

(c) True.

Suppose $A^T A = I$. We know
that I is invertible. By
Tut 4, Q3(c), A (and A^T)
are invertible, and hence
non-singular.

✓✓

Alternative solution using determinants:

Suppose $A^T A = I$.

Then $\det(A^T A) = \det(I)$.

Hence

$$\underbrace{\det(A^T) \cdot \det(A)}_{\det(A)} = \det(A^T A) = 1.$$

It follows that

$$(\det(A))^2 = 1,$$

and therefore,

$$\det(A) \neq 0,$$

which implies that A is non-singular.

(d) Converse is false.

Counterexample:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ is non-singular,}$$

$$\text{but } A^T A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\neq I$$

("almost any" non-singular
matrix serves as counterexample.)

$$\textcircled{7} \quad A = \begin{pmatrix} 2 & 0 & 0 \\ -2 & -4 & 2 \\ 2 & -1 & -1 \end{pmatrix}$$

$$(a) \quad 0 = \begin{vmatrix} 2-\lambda & 0 & 0 \\ -2 & -4-\lambda & 2 \\ 2 & -1 & -1-\lambda \end{vmatrix} \quad \checkmark$$

$$= (2-\lambda) \begin{vmatrix} -4-\lambda & 2 \\ -1 & -1-\lambda \end{vmatrix}$$

$$= (2-\lambda) [(-4-\lambda)(-1-\lambda) + 2] \quad \checkmark$$

$$= (2-\lambda) [4 + 4\lambda + \lambda + \lambda^2 + 2]$$

$$= (2-\lambda) (\lambda^2 + 5\lambda + 6)$$

$$= (2-\lambda) (\lambda+2) (\lambda+3)$$

Hence the eigenvalues are

$$\lambda = -3, \lambda = -2, \lambda = 2. \quad \checkmark$$

(b) Eigenvectors corresponding to $\lambda = -3$:

$$(A - \lambda I)\underline{x} = \underline{0}$$

$$A - (-3)I = \begin{pmatrix} 5 & 0 & 0 \\ -2 & -1 & 2 \\ 2 & -1 & 2 \end{pmatrix} \checkmark$$

$$\begin{pmatrix} 5 & 0 & 0 \\ -2 & -1 & 2 \\ 2 & -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \checkmark$$

One eigenvector was enough.

You may see directly that (for instance), $\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$ is a solution.

Otherwise, carry on with Gauss reduction:

$$\left(\begin{array}{ccc|c} 5 & 0 & 0 & 0 \\ -2 & -1 & 2 & 0 \\ 2 & -1 & 2 & 0 \end{array} \right) R_1 \rightarrow \frac{1}{5} R_1$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ -2 & -1 & 2 & 0 \\ 2 & -1 & 2 & 0 \end{array} \right) \begin{array}{l} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 2 & 0 \end{array} \right) R_3 \rightarrow R_3 - R_2$$

$$\begin{array}{c} x_1 \quad x_2 \quad x_3 \\ \left(\begin{array}{ccc|c} \textcircled{1} & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \end{array}$$

$$\underline{\underline{x_3 = t}}, \quad t \in \mathbb{R}$$

$$-x_2 + 2x_3 = 0$$

$$-x_2 + 2t = 0$$

$$\underline{\underline{x_2 = 2t}}$$

$$\underline{\underline{x_1 = 0}}$$

Hence eigenvectors :



$$\underline{\underline{\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} t}}, \quad t \in \mathbb{R}, \quad t \neq 0$$

(one eigenvector was sufficient!)

8 (a) If A and B are $n \times n$ matrices then

$$\det(AB) = \det(A) \cdot \det(B).$$

✓

(b) Since B is singular, there is an $\underline{x} \neq \underline{0}$ s.t. $B\underline{x} = \underline{0}$.

$$\text{Hence } A(B\underline{x}) = (AB)\underline{x} = \underline{0}$$

for some $\underline{x} \neq \underline{0}$.

Hence AB is singular. ✓

(c) Since AB is singular, $\det(AB) = 0$.

Since B is singular, $\det(A) \cdot \det(B)$

$$= \det(A) \cdot 0 = 0. \quad \checkmark$$

(d) Since B is non-singular, B is the product of elementary matrices (invertibility theorem). ✓

(e) line (5) ✓

~~(f) We apply "det(CE) = det(C) \cdot det(E)" of line (7)~~

(f) we apply

$$\text{"det}(CE) = \text{det}(C) \cdot \text{det}(E)\text{"}$$

of line (7) for

$$E = E_k \text{ and } C = AE_1 \dots E_{k-1}.$$

(g) By (**), we have that

$$E_3 E_2 E_1 A = \begin{pmatrix} 1 & 0 & -5 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Hence

$$\text{det}(E_3 E_2 E_1 A) = \begin{vmatrix} 1 & 0 & -5 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot 4 \cdot 1 = 4$$

It follows that

$$\underbrace{\text{det}(E_3)}_{-1} \cdot \underbrace{\text{det}(E_2)}_{-1/3} \cdot \underbrace{\text{det}(E_1)}_1 \cdot \text{det}(A) = 4.$$

$$\frac{1}{3} \cdot \text{det}(A) = 4$$

$$\underline{\underline{\text{det}(A) = 12}}$$

(h)

Fact The transpose of any elementary matrix is an elementary matrix.

Proof

(1) If E interchanges rows then
 $E^T = E$

(2) If E multiplies row i by K
 then $E^T = E$

(3) If E subtracts $K \cdot R_j$ from
 R_i then E^T subtracts $K \cdot R_i$
 from R_j .

□ (Fact)

✓✓

Let C be any $n \times n$ matrix and E
 be any $n \times n$ elementary matrix.
 Then

$$\begin{aligned} \det(CE) &= \det((CE)^T) \\ &= \det(E^T C^T) \\ &= \det(E^T) \cdot \det(C^T) \end{aligned}$$

(by (6) since E^T is elementary)

$$= \det(E) \cdot \det(C)$$

$$= \det(C) \cdot \det(E) \quad \checkmark \checkmark$$